

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR



DATA MINING

CS356

HOTS QUESTIONS

Submitted by : Group 4

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HOTS QUESTIONS

Q1. Compare Logistic Regression and Decision Tree for classification in the Plant-Grub dataset.

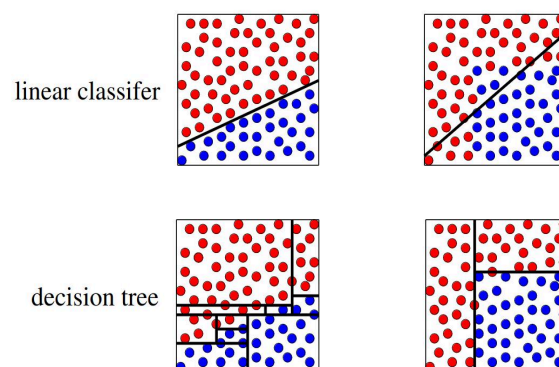
Answer:

Logistic Regression and Decision Tree are two fundamental classification algorithms used in data mining, but they differ significantly in their working principles and applicability. Logistic Regression is a statistical model that predicts the probability of a binary outcome using a sigmoid function. It assumes that there exists a linear relationship between the independent variables and the log-odds of the dependent variable. Due to this assumption, it performs well when the data is linearly separable and relationships between features are simple.

On the other hand, Decision Tree is a non-parametric supervised learning algorithm that works by recursively splitting the dataset into smaller subsets based on feature values. It creates a tree-like structure of decisions, where each node represents a feature and each branch represents a condition. Unlike Logistic Regression, it does not assume any linear relationship and can handle both linear and non-linear data effectively.

In the context of the Plant-Grub damage dataset, where multiple environmental and biological factors influence the outcome, Decision Tree may perform better due to its ability to capture complex feature interactions. However, it is more prone to overfitting, especially when the tree becomes very deep. Logistic Regression, although simpler, provides better generalization and interpretability.

Therefore, the choice between the two depends on the nature of the dataset. Logistic Regression is suitable for simple, interpretable models, whereas Decision Tree is better for capturing complex patterns.



Q2. Which algorithm is more suitable for non-linear data and why? (SVM, KNN, Logistic Regression)

Answer:

Aspect	Logistic Regression (LR)	Support Vector Machine (SVM)	K-Nearest Neighbors (KNN)
Type of Model	Linear model	Can be linear + non-linear (using kernels)	Non-parametric, instance-based
Decision Boundary	Linear (straight line)	Non-linear (with kernel trick)	Highly flexible, non-linear
Handling Non-linear Data	Poor performance	Very good (using RBF, polynomial kernels)	Good (adapts based on neighbors)
Working Principle	Uses sigmoid function to classify data	Finds optimal hyperplane with maximum margin	Classifies based on majority of nearest neighbors
Assumptions	Assumes linear relationship	No strict assumption (depends on kernel)	No assumption about data distribution
Flexibility	Low	High	High
Computation	Fast and efficient	Moderate (can be costly for large data)	Slow for large datasets (distance calculation)

Interpretability	High (easy to understand)	Low (complex mathematical model)	Moderate (depends on data)
Suitability for Dataset	Simple, linearly separable data	Complex, high-dimensional data	Small to medium, non-linear data
Example (Plant-Grub)	May fail if patterns are complex	Can capture complex environmental relationships	Works well if similar conditions give similar results

Q3. Explain why Decision Trees are prone to overfitting and how it can be controlled.

Answer:

Overfitting is a common problem in machine learning where a model learns not only the underlying patterns in the training data but also the noise. Decision Trees are particularly prone to overfitting due to their hierarchical and recursive nature of splitting data.

A Decision Tree works by selecting the best feature at each step to split the dataset in a way that maximizes information gain or minimizes impurity. As the tree grows deeper, it continues to create more specific rules that perfectly classify the training data. However, these rules may not generalize well to unseen data.

In the Plant-Grub dataset, overfitting may occur if the model starts capturing noise such as random variations in environmental conditions rather than actual patterns.

To control overfitting, several techniques can be used. Pruning is one of the most effective methods, where unnecessary branches are removed from the tree. Additionally, limiting the maximum depth of the tree, setting a minimum

number of samples required to split a node, and using ensemble methods like Random Forest can help reduce overfitting.

Thus, while Decision Trees are powerful, proper control techniques are necessary to ensure good generalization.

Q4. Discuss the impact of choosing different values of 'k' in KNN.

Answer:

K-Nearest Neighbors (KNN) is a simple yet effective classification algorithm that relies on the concept of similarity between data points. The parameter 'k' represents the number of nearest neighbors considered while making a prediction, and its choice significantly affects the model's performance.

When the value of k is very small, such as $k=1$, the model becomes highly sensitive to noise and small variations in the dataset. This leads to overfitting, where the model performs well on training data but poorly on test data.

On the other hand, when k is very large, the model considers too many neighbors, including those that may belong to other classes. This can smooth out the decision boundary excessively and lead to underfitting, where important patterns in the data are ignored.

In the context of the Plant-Grub dataset, selecting an optimal value of k is important to balance bias and variance. This is usually done using cross-validation, where different values of k are tested and the one with the best performance is selected.

Q5. Explain the importance of feature selection in this project.

Answer:

Feature selection is an important step in any machine learning project as it involves identifying the most relevant input variables that contribute to the prediction of the target variable. In the Plant-Grub damage dataset, features may include environmental conditions, soil properties, and biological factors.

Not all features contribute equally to the model's performance. Some features may be irrelevant or redundant, which can increase model complexity and reduce accuracy. By selecting only the important features, the model becomes simpler, faster, and more accurate.

Feature importance can be determined using techniques such as analyzing coefficients in Logistic Regression or using feature importance scores in Decision Trees. This helps in understanding which factors have the greatest impact on plant damage.

Additionally, feature selection improves interpretability, which is crucial in real-world applications like agriculture, where understanding the cause of predictions is as important as the prediction itself.

Q6. Why is accuracy not always a reliable metric? Explain with respect to this dataset.

Answer:

Accuracy is one of the most commonly used evaluation metrics in classification problems. It measures the proportion of correct predictions made by the model. However, it is not always a reliable metric, especially in cases of imbalanced datasets.

In the Plant-Grub dataset, if one class (e.g., “no damage”) is significantly more frequent than the other class (e.g., “damage”), a model may achieve high accuracy simply by predicting the majority class every time. This creates a false impression of good performance while failing to correctly identify the minority class.

To address this issue, other evaluation metrics such as precision, recall, and F1-score are used. Precision measures how many predicted positives are actually correct, while recall measures how many actual positives are correctly identified. F1-score provides a balance between precision and recall.

Therefore, relying only on accuracy can be misleading, and multiple metrics should be considered for a comprehensive evaluation.

Q7. Discuss the importance of normalization in machine learning algorithms.

Answer:

Normalization is the process of scaling input features to a common range, usually between 0 and 1 or -1 and 1. It plays a crucial role in machine learning algorithms that rely on distance calculations, such as KNN and SVM.

In these algorithms, the distance between data points is used to determine similarity. If features are not normalized, those with larger values can dominate the distance calculation, leading to biased results.

For example, in the Plant-Grub dataset, if one feature is measured in large units (e.g., rainfall in mm) and another in small units (e.g., temperature in degrees), the algorithm may give more importance to rainfall simply because of its larger scale.

However, normalization is not required for Decision Trees because they do not rely on distance measures. They split data based on feature values and thresholds.

Thus, normalization is essential for certain algorithms to ensure fair and accurate predictions

Q8. Explain the bias-variance tradeoff with respect to different models applied on this dataset.

Answer:

The bias-variance tradeoff is a fundamental concept in machine learning that explains the balance between underfitting and overfitting. **Bias** refers to the error due to overly simplistic assumptions in the model, while **variance** refers to the error due to excessive sensitivity to training data.

In the Plant-Grub dataset, models like **Logistic Regression** typically have high bias and low variance, meaning they may underfit complex relationships in the data. On the other hand, models like **Decision Trees** have low bias but high variance, as they can capture intricate patterns but may overfit the training data.

Algorithms like **Random Forest** and **SVM** attempt to balance this tradeoff. Random Forest reduces variance by averaging multiple trees, while SVM controls complexity using regularization and kernel functions.

Thus, selecting the right model involves balancing bias and variance to achieve optimal generalization performance.

Q9. How would you evaluate the performance of a classification model beyond traditional metrics in this agricultural context?

Answer:

In real-world agricultural applications, evaluating a model solely based on metrics like accuracy or F1-score is insufficient. It is important to consider the **practical impact of predictions**. For example, misclassifying a diseased plant as healthy may result in severe crop loss, making false negatives more critical than false positives.

Therefore, domain-specific evaluation metrics such as **cost-sensitive analysis** should be used, where different types of errors are assigned different weights. Additionally, **ROC-AUC** can be used to measure the model's ability to distinguish between classes across different thresholds.

Another important aspect is **model interpretability**, as farmers need to trust and understand the predictions. Hence, evaluation should also include explainability techniques and real-world validation through field testing.

Q10. Discuss how temporal and spatial factors could influence the predictive modeling in the grub damage dataset.

Answer:

Temporal and spatial factors play a crucial role in agricultural datasets. Temporal factors such as **year, season, and climate changes** can significantly influence pest behavior and crop health. Similarly, spatial factors such as **geographical zones, soil type, and irrigation patterns** affect the distribution of grub damage.

Ignoring these factors can lead to models that fail to generalize across different regions or time periods. For example, a model trained on data from one year may not perform well in another year due to climate variations.

To address this, techniques such as **time-series analysis, spatial clustering**, or inclusion of location-based features can be used. Models should also be validated using data from different time periods and regions to ensure robustness.

Q11. How can ensemble learning improve prediction accuracy in this project, and what are its limitations?

Answer:

Ensemble learning combines multiple models to improve overall prediction performance. In the Plant-Grub dataset, techniques like **Random Forest** and **Gradient Boosting** can significantly enhance accuracy by reducing variance and capturing complex patterns.

Random Forest works by averaging predictions from multiple decision trees, thereby reducing overfitting. Gradient Boosting builds models sequentially, focusing on correcting errors made by previous models.

However, ensemble methods have limitations. They are computationally expensive, harder to interpret, and may require careful tuning of hyperparameters. In agricultural applications, where interpretability is important for decision-making, this can be a drawback.

Q12. How would you handle concept drift in the Plant-Grub dataset when deploying a long-term predictive model?

Answer:

Concept drift refers to the change in the underlying data distribution over time, which can significantly affect the performance of machine learning models. In the context of the Plant-Grub dataset, factors such as climate change, evolving pest behavior, soil degradation, and changes in agricultural practices can cause the relationship between input features and target variables to shift over time.

If a model is trained on historical data and deployed without updates, its accuracy may degrade as new patterns emerge. For example, a pest that was previously active only in a specific season may start appearing throughout the year due to environmental changes.

To handle concept drift, several strategies can be applied. One approach is **incremental learning**, where the model is continuously updated with new incoming data. Another method is **periodic retraining**, where the model is retrained at regular intervals using recent data. Additionally, **drift detection**

techniques can be used to identify when significant changes occur in data distribution.

Maintaining a feedback loop from real-world predictions and monitoring model performance over time is also essential. This ensures that the model remains relevant, accurate, and reliable in dynamic agricultural environments.

THANK YOU